

Keep the Fish Healthy with this IoT Based Aquarium

Fish living in aquariums need constant monitoring. This article outlines the development of an IoT based monitoring system for aquariums, which eliminates the need for human intervention to check the water quality in them.

Beautiful and delicate marine life creatures require a lot of care. Misconceptions such as fish can survive without food or can live with low water quality in an aquarium can endanger them. Fish require sufficient levels of nutrients in the water they swim in. In fact, the oxygen content in the water is vital. Dissolved oxygen (DO) is a measure of the amount of oxygen present in the water. Most fish need DO content in the range of 5-6ppm in a water tank, even if some can manage at 4ppm. Anything

lesser is dangerous. The pH value is also an important aspect in determining the quality of the water. This value needs to be monitored frequently to maintain a healthy environment for the fish in an aquarium.

Let us now use the Internet of Things (IoT) to run a framework that can detect and observe oxygen levels, temperature, pH of the water, the water level, and so on, in an aquarium by using a few sensors. Keeping up

with the right conditions in the water is extremely hard manually. Automation can significantly diminish the fatality rate and help make an aquarium a healthy environment for fish to live in.

Making an aquarium intelligent

The system we are developing aims to couple existing fish automation systems such as water-quality monitoring with automating the tank cleaning process, the oxygenator, the temperature filter, pH value sensor and the DO sensor. A temperature sensor is

